# YUXUAN HOU

Shenzhen, China | 12413104@mail.sustech.edu.cn | tsawke.github.io
Google Scholar | Personal website



## Education

**Southern University of Science and Technology (SUSTech)** — **Expected Jun 2028**

*B.Eng. in Computer Science and Technology — Turing Class (Honors Program)* — *Shenzhen, China*

**Academic-year GPA (2025–2026): 3.86/4.00**; ranked top 10% in comprehensive evaluation

**Honors Coursework:** Discrete Mathematics; Data Structures and Algorithm Analysis; Algorithm Design and Analysis; Principles of Database Systems; Artificial Intelligence.

**Other Completed Coursework:** Computer Organization; Computer Vision; Natural Language Processing; Digital Logic; Introduction to Mathematical Logic; Introduction to Computer Programming; Computational Ethics; Group Projects I.

**In Progress:** Operating Systems (Honors); Compiler Principles; Computer Networks; Advanced Computer Programming; Probability and Statistics for Engineering.

**Lab:** **Advanced Intelligent Mobile Systems (AIMS) Lab**, Department of Computer Science and Engineering, SUSTech
**Advisors:** **Jin Zhang** (Associate Professor) and **Linshan Jiang** (Assistant Professor)

## Publications & Manuscripts

[1] **Yuxuan Hou** et al. *SNN-STEM: Towards High-Performance Aerial Object Detection with Frequency-Enhanced Spiking Neural Networks*.
Manuscript in preparation; planned submission to **CVPR 2027**.

[2] **Yuxuan Hou**, Di Yu, Linshan Jiang, and Jin Zhang. *Understanding Spike Encoding across IoT Sensing Modalities: An Empirical Study*.
Manuscript submitted to **SenSys 2027**.

[3] **Yuxuan Hou** and Linshan Jiang. *LADS: Lease-Aware Delivery Scheduling of Tool Results for Language Model Agents*.
Manuscript submitted to **The Web Conference (WWW) 2027**.

[4] **Yuxuan Hou**, Jiao Li, Linshan Jiang, and Jin Zhang. *Oto-Meal: Earable Sensing with PPG and IMU for Personalized Meal Awareness*.
**WellComp 2026 (UbiComp/ISWC 2026 Workshop)** — Computing for Well-being.
Accepted and presented, October 2026; forthcoming in **UbiComp Companion '26**. DOI: 10.1145/3798063.3841767. arXiv:2608.16678.

## Research Experience

### Neuromorphic Computing & Sensing — Ongoing

**SNN-STEM: Towards High-Performance Aerial Object Detection with Frequency-Enhanced Spiking Neural Networks** — **Ongoing**

*First Author* — *Planned* **CVPR 2027**
Advancing spiking-neural-network models and evaluation for ground-object detection in aerial imagery.

**Understanding Spike Encoding across IoT Sensing Modalities: An Empirical Study** — **Submitted**

*First Author* — **SenSys 2027**

- Led study design, implementation and manuscript writing, with controlled comparisons across **six tasks and four modalities**: wireless, inertial, electromyographic and tactile signals; evaluated accuracy, inference-pipeline latency and measured power-rail energy on **NVIDIA Jetson AGX Orin**.

**Oto-Meal: Earable Sensing with PPG and IMU for Personalized Meal Awareness** — **Sep 2025–Present**

*First Author* — *Wearable sensing*
First-author paper accepted to and presented at **WellComp 2026 (UbiComp/ISWC 2026 Workshop)**.
**Project Lead** — 2026 National Undergraduate Innovation and Entrepreneurship Training Program (**National Level**; CNY 25,000)
**Project Lead** — 2026 SUSTech "Climbing Program" Project (CNY 15,000; completed)

- Built an audio- and image-free earable prototype using photoplethysmography (PPG) and inertial sensing; evaluated a pooled neural classifier separately from within-user memory matching on **538 recordings and 7,028 windows from 7 participants**.
- Extended follow-up work through controlled cross-user transfer studies and a browser-based acquisition and video-review prototype for future full-meal data collection.

## Research Experience (continued)

**LADS: Lease-Aware Delivery Scheduling of Tool Results for Language Model Agents** **2026**

*First Author* *Submitted*

- Developed tool-result scheduling for concurrent language-model agents and a real-inference evaluation system.
- Reduced mean complete-workflow latency by **44.6%** on controlled bursts (5 input seeds) and **42.4%** on Web-derived burst replay (3 input seeds), versus ReAct with first-in, first-out delivery, using the same model and agent policy.

**3D Acoustic Field Reconstruction and Adaptive Acoustic Modeling** **May 2026–Present**

*Acoustic Modeling & Simulation* *In preparation*

- Developed simulation-based evaluation of acoustic field reconstruction under sparse, imperfect measurements, covering object-response reuse, receiver-position calibration and room-response prediction; prepared protocols for future real-room measurements.

**Related computer vision course project:** Team Leader, four-person Object-Level 3D Gaussian Splatting project (Spring 2026); built reconstruction, segmentation and editing pipeline, registering 213/213 frames with 0.71-pixel reprojection error.

**RydbergTrack: Motion Sensing with a Rydberg Atomic Quantum Receiver** (Spring semester 2026; Team Leader, three-person project): built a reproducible MATLAB pipeline for quantum-receiver signal analysis; processed 150 million raw samples and separated 3 static from 12 motion recordings within the project dataset.

## Engineering Projects

**SUS-Core-Agent** (2026; Developer): built a VS Code extension for Vivado simulation, synthesis and bitstream workflows, with source/log-aware AI tutoring, authenticated accounts and per-user records.
**ZhirenHub**: independently developed a WeChat roommate-matching prototype with eligibility verification, preference matching, team invitations, dormitory quotas and administration using Taro, React and CloudBase.
**Dormitory Safety and Hygiene Rating Query**: independently developed a WeChat mini program for students to query dormitory assessment results.
**LoongArch CPU-Based SoC** (Team Leader; three-person course project): led integration of single-cycle/pipelined execution, UART debugging, FPGA peripherals, simulation and differential-testing workflows.
**FPGA Matrix Calculator with CNN Accelerator** (Team Leader; three-person course project): designed top-level architecture and the control FSM; built Web Serial verification, build automation and recovery mechanisms.

## Technical Skills

**Programming:** Python, C/C++, MATLAB, Java, JavaScript, Verilog
**ML & Scientific Computing:** PyTorch, scikit-learn, NumPy, SciPy; SNNs, signal processing, numerical optimization
**Tools:** CUDA, COLMAP, Vivado, Node.js, Git, Linux, LaTeX; inference scheduling, reproducible evaluation and test automation
**Languages:** Chinese; English — Duolingo English Test: **120**, minimum subscore **115** (Oct 2026); CET-6: **570**

## Selected Honors & Awards

**2025** **National First Prize**, Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM)
**2025** **Silver Medal (Rank 4)**, International Collegiate Programming Contest (ICPC)
**2025** **Silver Medal**, China Collegiate Programming Contest (CCPC)
**2025** **Gold Medal**, Guangdong Collegiate Programming Contest
**2026** **International First Prize (1st Place)**, Application Development Division Finals, 1st Hong Kong International AIGC Cultural Digital Content Creation Competition
**2025** Provincial First Prize, 16th Lanqiao Cup, Guangdong; C/C++ Programming, University Group A
**2025/26** SUSTech **Outstanding Student Leader (2025)**; **Exemplary Student Leader (2026)**
**2025/26** SUSTech University Scholarship (2025 and 2026); Zhiren College Motto Scholarship “Rixin” (2025)
**2026** Undergraduate Talent Development Excellence Program Grant (2024 Cohort; CNY 6,000)
**2025/26** SUSTech Outstanding Communist Youth League Cadre
**2026** Second Prize, CCF Service Computing Innovation Competition / Sangfor CoStrict AI-Native Innovation Competition; *GeoAgent* (team project)

**Earlier honors:** Provincial Rank 9, National Olympiad in Informatics (NOI) Joint Provincial Selection; First Prizes in the NOI Spring Test and CCF CSP-S (senior-level certification); Outstanding Student, Changchun.

## Leadership & Service

**President, Student Union | Chair, Student Development Committee:** lead organizational planning, team coordination, recruitment, student activities, member development, leadership handovers, and operating procedures.

**Head, Zhiren College Peer Mentor Team:** lead 15 mentors serving 231 incoming students across 5 classes and outreach covering 25 provinces; organize onboarding, orientation, and ongoing support.
**Collective honors:** SUSTech Outstanding Student Union; First Prize, Zhiren College Outstanding Student Organization (2025–2026).
**Volunteer service: 87.17 verified hours** (September 2026), including campus events and accessibility initiatives.
**Other roles:** Turing Class President; 3D Manufacturing Club Vice President; competitive programming, supercomputing, rowing, emcee, and exhibition hall docent teams.